

CMOS Inverter problems

① symmetric baseline

$$V_{DD} = 1.8 \text{ V} ; |V_{tp}| = 0.4 \text{ V}$$

$$V_{tn} = 0.4 \text{ V} ; k_n = k_p = \frac{200}{200} \mu\text{A/V}^2 \quad C_L = 50 \text{ fF}$$

$$V_{OH} \Rightarrow V_{DD} = 1.8 \text{ V}$$

also

$$V_{OL} \Rightarrow 0 \text{ V}$$

To find switching threshold (V_M)

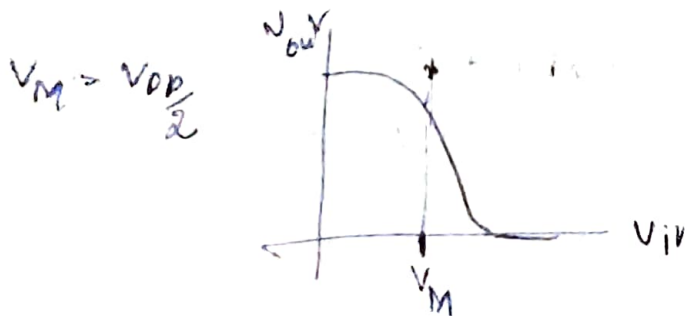
$$\text{we need } \sqrt{\frac{k_n}{k_p}} = 1 //$$

$$V_M = \frac{V_{DD} - |V_{tp}| + V_{tn}}{1+1} = \frac{1.8 - 0.4 + 0.4}{2} = 0.9 \text{ V}$$

For propagation delay,

$$I_{\text{sat}, n} = \frac{200 \times 10^{-6}}{2} (1.8 - 0.4)^2 = 100 \times 10^{-6} \times 1.96 = 196 \mu\text{A}$$

$$t_{PHL} = \frac{C_L (V_{DD}/2)}{I_{\text{sat}, n}} = \frac{50 \times 10^{-15} \times 0.9}{196 \times 10^{-6}} = 229 \text{ ps} //$$



② $V_{DD} = 2.5V$ $V_{tn} = 0.5V$, $|V_{tp}| = 0.5V$, $k_n = 120 \mu A/V^2$
 $k_p = 40 \mu A/V^2$ $C_L = 100 fF$

$V_{OH} = 2.5V$; $V_{OL} = 0V$

for V_M ; $\sqrt{\frac{k_n}{k_p}} = \sqrt{3} = 1.732$

$V_M < \frac{V_{DD}}{2}$

$V_M = \frac{2.5 - 0.5 + 0.5(1.732)}{1 + 1.732} = 1.249V$ or $\sqrt{\frac{k_n}{k_p}} > 1$

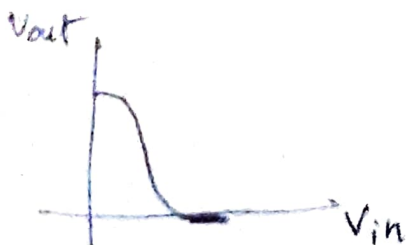
$I_{sat,n} = \frac{120 \times 10^{-6}}{2} (2.5 - 0.5)^2 = 240 \times 10^{-6} A$

$t_{PHL} = \frac{100 \times 10^{-15} \times 1.25}{240 \times 10^{-6}} = 521 pS$

$V_M < \frac{V_{DD}}{2} \rightarrow$ curve shifts ~~right~~ left, strong NMOS

$I_{sat,p} = \frac{40 \times 10^{-6}}{2} (2.5 - 0.5)^2 \Rightarrow 80 \times 10^{-6} A$

$t_{PLH} \Rightarrow \frac{100 \times 10^{-15} \times 1.25}{80 \times 10^{-6}} = 1.56 ns$



$$(3) \quad V_{DD} = 1.2V \quad V_{th} = 0.3V, \quad |V_{tp}| = 0.3V, \quad k_n = 300 \mu A/V^2$$

$$k_p = 150 \mu A/V^2 \quad C_L = 20 fF$$

$$V_{OH} = 1.2$$

$$V_{OL} = 0V$$

$$\sqrt{\frac{k_n}{k_p}} = \sqrt{2} = 1.414$$

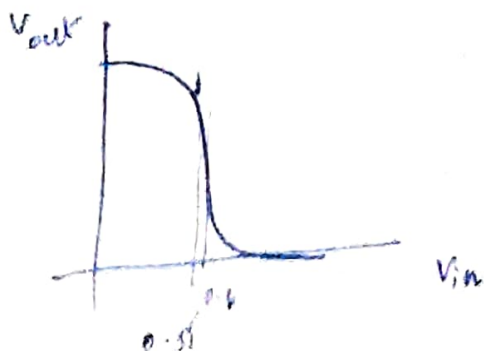
$$V_M = \frac{1.2 - 0.3 + 0.3(1.414)}{1 + 1.414} = 0.548V$$

$$I_{sat,n} = \frac{300 \mu}{2} (1.2 - 0.3)^2 = 121.5 \mu A$$

$$t_{PHL} = \frac{20 fF \times 0.6}{121.5 \mu A} = 98.7 \times 10^{-12} s$$

$$I_{sat,p} = \frac{150 \mu}{2} (1.2 - 0.3)^2 = 60.75 \mu A$$

$$t_{pLH} = 197.5 ps \text{ (approx)}$$



$$\textcircled{H} \quad V_{DD} = 3.3 \text{ V} \quad V_{th} = 0.7 \text{ V} \quad |V_{tp}| = 0.9 \text{ V} \quad k_n = 150 \mu\text{A/V}^2$$

$$k_p = 150 \mu\text{A/V}^2 \quad C_L = 80 \text{ fF}$$

$$V_{OH} = 3.3 \text{ V}$$

$$V_{OL} = 0 \text{ V}$$

$$V_M = \frac{3.3 - 0.9 + 0.7}{2} = 1.55 \text{ V}$$

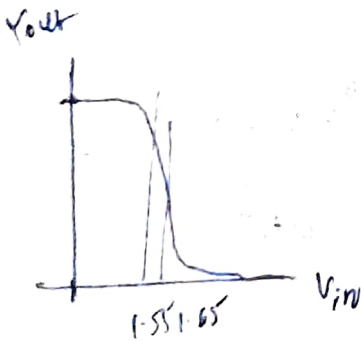
$$\sqrt{\frac{k_n}{k_p}} = 1$$

$$I_{sat,n} = \frac{150 \mu (3.3 - 0.7)^2}{2} = 507 \mu\text{A}$$

$$t_{pHL} \approx 260 \text{ ps}$$

$$I_{sat,p} = \frac{150 \mu (3.3 - 0.9)^2}{2} = 432 \mu\text{A}$$

$$t_{pLH} \approx 305 \text{ ps}$$



$$\textcircled{5} \quad V_{DD} = 1.0 \text{ V} \quad |V_{tp}| = 0.25 \text{ V}$$

$$V_{tn} = 0.25 \text{ V}$$

$$k_n = 400 \mu\text{A/V}^2$$

$$k_p = 200 \mu\text{A/V}^2 \quad C_L = 250 \text{ fF}$$

$$V_{OH} = 1 \text{ V}$$

$$V_{OL} = 0 \text{ V}$$

$$\sqrt{\frac{k_n}{k_p}} = \sqrt{2} = 1.414$$

$$V_M = \frac{1 - 0.25 + 0.25(1.414)}{1 + 1.414} = 0.457 \text{ V}$$

$$I_{\text{sat}, n} = \frac{400 \times 10^{-6}}{2} (1 - 0.25)^2 = 112.5 \mu\text{A}$$

$$t_{pHL} = \frac{250 \text{ fF} \times 0.5 \text{ V}}{112.5 \mu\text{A}} = 1.11 \text{ ns}$$

$$I_{\text{sat}, p} = \frac{200 \times 10^{-6}}{2} (1 - 0.25)^2 = 100 \times 10^{-6} \times 0.5625$$

$$= 56.25 \mu\text{A}$$

$$t_{pLH} = 2.22 \text{ ns}$$